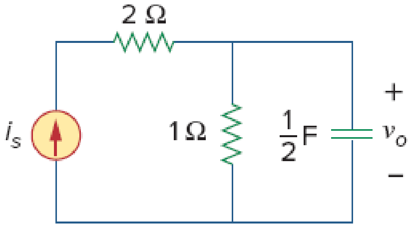
Chapter 18, Problem 47P

Problem

Find the voltage *vo*(*t*) in the circuit of Fig. 18.49. Let *is*(*t*) = 8*e*−*tu*(*t*) A.

Figure 18.49



Step-by-step solution

Step 1 of 4

Consider the following circuit diagram:

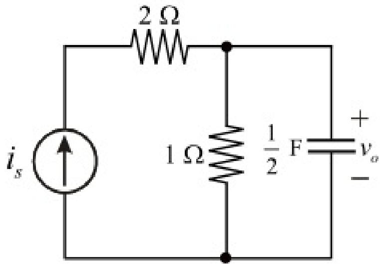


Figure 1

Step 2 of 4

Consider the current source in the circuit, $i_s = 8e^{-t}u(t)$.

Convert the time domain function i_s into the frequency domain using Fourier transform pair.

$$\begin{aligned} I_s &= \mathcal{F}(i_s) \\ &= \mathcal{F}\left[8e^{-t}u(t)\right] \\ &= 8\mathcal{F}\left[e^{-t}u(t)\right] \qquad \left(\because\ e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a+j\omega}\right) \\ &= \frac{8}{j\omega+1} \end{aligned}$$

Write the frequency domain function for capacitor value.

$$\frac{1}{2}\text{ F} \longleftrightarrow \frac{1}{j\omega\left(\frac{1}{2}\right)}$$

Draw the Figure 1 in Frequency domain.

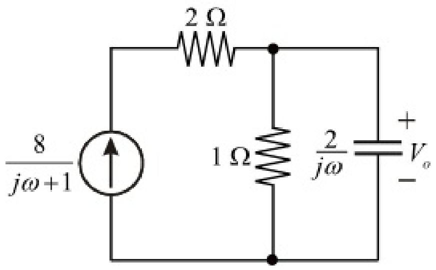


Figure 2

Step 3 of 4

Calculate the output voltage in frequency domain from Figure 2.

$$\begin{aligned} V_o &= \frac{2}{j\omega}(I_o) \\ &= \frac{2}{j\omega}\left(\frac{(1)I_s}{1+\frac{2}{j\omega}}\right) \\ &= \frac{2}{j\omega}\left(\frac{(j\omega)I_s}{j\omega+2}\right) \\ &= 2\left(\frac{1}{j\omega+2}\right)\left(\frac{8}{j\omega+1}\right) \\ &= \frac{16}{(j\omega+2)(j\omega+1)} \end{aligned}$$

Solve the V_o using the residual partial fraction rule.

$$\begin{aligned} V_o &= \frac{16}{(j\omega+2)(j\omega+1)} \\ &= \frac{A}{(j\omega+2)} + \frac{B}{(j\omega+1)} \end{aligned}$$

Here *A* and *B* are constants.

Step 4 of 4

Calculate the constant value *A*, using the residue method.

$$\begin{aligned}A &= (j\omega+2)V\big|_{j\omega=-2}\\ &= (j\omega+2)\left(\frac{16}{(j\omega+1)(2+j\omega)}\right)\Bigg|_{j\omega=-2}\\ &= \left(\frac{16}{(-2+1)}\right)\\ &= -16\end{aligned}$$

Calculate the constant value *B* using the residue method.

$$\begin{aligned}B &= (j\omega+1)V_o\big|_{j\omega=-1}\\ &= (j\omega+1)\left(\frac{16}{(j\omega+2)(j\omega+1)}\right)\Bigg|_{j\omega=-1}\\ &= \left(\frac{16}{(2-1)}\right)\\ &= 16\end{aligned}$$

Substitute obtained *A* and *B* values in equation V_o.

$$V_o = \frac{16}{(j\omega+1)} + \frac{-16}{(j\omega+2)}$$

Apply the inverse Fourier transform to find $v(t)$.

$$\begin{aligned}v(t) &= \mathcal{F}^{-1}(V_o)\\ &= \mathcal{F}^{-1}\left[\frac{16}{(j\omega+1)} + \frac{-16}{(j\omega+2)}\right]\\ &= \mathcal{F}^{-1}\left(\frac{16}{(j\omega+1)}\right) - \mathcal{F}^{-1}\left(\frac{16}{(j\omega+2)}\right) \qquad \left(\because\ e^{-at}u(t) \xleftrightarrow{\mathcal{F}} \frac{1}{a+j\omega}\right)\\ &= 16e^{-t}u(t) - 16e^{-2t}u(t)\\ &= 16\left(e^{-t} - e^{-2t}\right)u(t)\ \text{V}\end{aligned}$$

Thus, the output voltage $v(t)$ is $\boxed{16\left(e^{-t}-e^{-2t}\right)u(t)\ \text{V}}$.